AutoBayes brms-parity loop: Tier D review

Status at the Tier D boundary, prepared for Andrew

17 October 2026

This is the Tier D boundary review, in the same form as the earlier ones. All eight Tier D features are on `main`, together with the cumulative probit link that D04 had to withhold until a 50-digit reference of it existed. Each feature was merged as `andrew-dou` on local verification, as you allowed: GitHub Actions still refuses to start any job on your account ("recent account payments have failed or your spending limit needs to be increased", checked again on the last merge). Section 2 lists two places where brms itself is wrong or inexact, and Section 7 is the part that needs you.

1 What was built

Table 1 lists the features.
AutoBayes now covers 30 of the 45 response families brms 2.23 documents.[1] The fifteen it lacks are `asym_laplace`, `cox`, `dirichlet`, `dirichlet_multinomial`, `frechet`, `gen_extreme_value`, `hurdle_cumulative`, `inverse.gaussian`, `logistic_normal`, `multinomial`, `von_mises`, `wiener`, `xbeta`, `zero_inflated_beta_binomial` and `zero_one_inflated_beta`. As before, every agent wrote from brms's documentation, the published densities and papers, and brms's outputs on probe inputs, never from brms's or loo's GPL source.

How close. Measured with the gates' own comparisons on `main` after the last merge (`0fb9649`): all 133 ungated cases pass, 35 of them new in Tier D. Over the Tier D cases the worst relative errors are 1.3×10^{-9} in G2, 9.7×10^{-16} in G3, 5.3×10^{-15} in predictions (`hurdle_negbinomial_varint`), 1.9×10^{-15} between the JAX and PyTensor densities, and 1.1×10^{-14} in the post-processing tables (`gauss_loo_compare_x`); G1's designs are exact. Their log densities are within 5×10^{-16} of brms's except `gauss_horseshoe`, at 1.3×10^{-10}: PyMC's float32 $\log 2$ in its six half-Student-t scales (D103). The gradients above 10^{-12} are the precision and shape gradients of `beta_binomial_varint` (1.3×10^{-9}), `hurdle_gamma_varint` (9.4×10^{-10}) and `weibull_varint` (9.2×10^{-10}), which go through PyTensor's digamma (Section 2). Over all 133 cases the worst G2 error is still Tier C's 7.2×10^{-9} (`gamma_distributional_shape`'s gradient). Every value is inside the unchanged tolerances: 10^{-8} for log densities, 10^{-7} for gradients, 10^{-9} for predictions.

2 What brms gets wrong

- **`hurdle_negbinomial` response draws (D86).** brms 2.23's `posterior_predict()` does not draw the positive part from the zero-truncated negative binomial. At a fixed point ($\mu = 0.5$, shape 0.5, $hu = 0.05$, a fixed-parameter fit, 1.9×10^5 positive draws) 81.6% of its positive draws

[1] The count comes from `brmsfamily()`'s documentation plus `Gamma` and `negbinomial2`, checked by calling `brmsfamily()` on each name. AutoBayes registers 31 family names; `mixture` is not one of brms's families.

Feature	PR	What a user can now write	Frozen brms cases it passes	Diverg.
D05 model comparison	#58	`ab.waic`, `ab.loo_compare({...})`, `ab.model_weights(results, "loo"|"waic"|"pseudobma")`	`gauss_loo_compare_x`, `_xw`	D92–D95
D08 grouping terms	#59	`(1 | mm(g, g2, weights = cbind(w1, w2)))`, `(1 | gr(g, by = gby))`, a correlation ID shared with `sigma`'s formula	`gauss_mm`, `gauss_gr_by`, `gauss_shared_id`	D104–D107
D01 continuous families	#60	`ab.weibull`, `ab.exponential`, `ab.skew_normal`, `ab.shifted_lognormal` (`ndt`, brms's `Uniform(0, min_Y)`), `ab.exgaussian`	five `*_varint`	D76–D79
D06 hypothesis tests and effects	#61	`ab.hypothesis(m, fit, "x - w > 0")` (own parser, never evaluated), `ab.conditional_effects(m, fit, ["x"])`	`gauss_hypothesis_effects`, `bernoulli_...`	D96–D99
D02 discrete families	#62	`ab.beta_binomial` (trials, `phi`), `ab.geometric`, `ab.negbinomial2` (`sigma` = 1/shape)	three `*_varint`	D80–D82
D07 shrinkage priors	#63	`Priors(b=ab.Horseshoe(df, scale_global, ...))`, `Priors(b=ab.R2D2(mean_R2, prec_R2, cons_D2))`	`gauss_horseshoe`, `gauss_r2d2`	D100–D103
D04 ordinal models	#64, #66	`ab.sratio`, `ab.cratio`, `ab.acat`; `cs(x)`; `ab.cumulative(..., link="probit")`	four `*_varint`, `acat_cs`, five new	D88, D89
D03 zero-inflated and hurdle	#65	`ab.zero_inflated_negbinomial`, `_binomial`, `_beta`, `ab.hurdle_negbinomial`, `_gamma`, `_lognormal`; `weights()` for all of them	eight cases	D84, D86

Table 1: The Tier D features, in merge order. #56 (mixture `log_lik`, a `describe()` order) and #57 (the 30 Tier D fixtures) merged before them.

are 1 where the distribution gives 60.4%, and their mean is 1.27 where it is 1.71; at $\mu = 3$, shape 2 the means are 3.15 and 3.57. brms's own `posterior_epred()` reports the correct mean (1.6218 $= 0.95 \times 1.7071$), so its draws contradict its expected value. D03 found this and recorded it in `tests/zero_hurdle_probes`; I reproduced it independently, with another seed, against rejection sampling from R's own `rnbinom`. AutoBayes draws exactly (the Gamma–Poisson rate given a positive count in closed form, then a zero-truncated Poisson), checked against the mass function for shapes from 10^{-12} to 10^{8}. I am fairly confident this is a bug in brms rather than a choice, since the draws disagree with brms's own expected value; I have not checked whether a later brms fixes it.

- **Stan's `ordered_probit_lpmf` gradient (D88).** On `cumulative_probit_varint` brms's log density equals the 50-digit value to 3.6×10^{-16}, but its gradient is off by up to 6.5×10^{-5} relative, at 173 of the 176 entries (with and without the Jacobian). `oracle/reference.py` now implements the cumulative probit model (Stan's ordered transform and its Jacobian, each row's Normal mass from the tail it lies in), and the case is accepted under P6 against it: AutoBayes is within 4×10^{-15} of the reference.

Two numerical limits are ours, not brms's, and are documented: PyMC's `BetaBinomial` loses about $\epsilon\,\phi \log \phi$ as the precision grows (1.4×10^{-7} at $\phi = 10^{8}$, where Stan's is within 9×10^{-9}; D82), and PyTensor's digamma is about 1.5×10^{-10} off for arguments between 1 and 12, which reaches the shape gradients on the PyTensor path (9.2×10^{-10} on `weibull_varint`, 9.4×10^{-10} on `hurdle_gamma_varint`, both inside the 10^{-7} tolerance). JAX's digamma is exact, and NumPyro samples on JAX.

3 What combining the features found

The eight features were written against `main`s that lacked one another. I merged them one at a time, rebasing each onto the last, re-running the gate tool on the tree that would merge and the whole non-sampling suite: 3,061 tests passed after D01, 3,174 after D06, 3,280 after D02, 3,345 after D07, 3,487 after D04, 3,725 after D03 and 3,734 after the probit link. What the combinations turned up:

- D06's post-processing refused the new ordinal families' thresholds correctly, but its tests listed the refusals they expected, and the monotonic-term test listed the families that refuse `mo()`; both were extended for D04.
- D04's own suite had never run (its agent was stopped by your pause). Running it found a test that used `sratio` as an example of an unregistered family, and an error message that called every ordinal family "the cumulative family".
- Offering the cumulative probit link made the `link=` refusal claim that the cumulative family takes the cloglog link; it now lists each multi-link family with its own links.
- `ab.log_lik`, `ab.bayes_R2` and `ab.loo` raised an `AttributeError` on a multivariate model; they now refuse it by name (D06).

The pause. When you paused the loop, four agents stopped mid-way. D02, D04 and D07 had finished implementing and were running their final suites, so I committed their work as it stood and verified it myself, re-taking the snapshot of every earlier case model (unchanged in all three). D03 was mid-implementation; a fresh agent reviewed the uncommitted work before building on it and found four real problems: a sampler that hung for over 40 minutes at shape 10^{-300}, an underflow that voided the whole zero-inflated Beta likelihood at $|\eta| > 745$, a $+\infty$ log density at an observed $y = 1$ where brms gives $-\infty$, and a divergence row that paraphrased brms's source, now replaced

by output-based evidence.

4 Where AutoBayes knowingly differs from brms (Tier D rows)

	area	what differs	status
D76–D79	continuous families	other links; formulas on `shape`, `alpha`, `beta`, `ndt`; addition, special and autocorrelation terms; multivariate and mixture use	kept
D80–D82	discrete families	the same contexts; beta-binomial precision at very large ϕ	kept / documented
D84, D86	zero-inflated, hurdle	other links, formulas on `zi`/`hu`, `cens()`/`trunc()`, special terms; brms's hurdle NB draws are wrong, ours exact	kept / documented
D88, D89	ordinal	cumulative probit accepted on P6's reference; other links, `disc`, `mo()`, other forms of `cs()`	accepted / kept
D92–D95	model comparison	stacking, BMA, `kfold`, the Bayesian bootstrap; mixed criteria refused	kept
D96–D99	`hypothesis()`, effects	Savage–Dickey ratios (need prior draws), other classes, interaction effects, most options	kept
D100–D103	shrinkage priors	other classes and formulas, `par_ratio`, `autoscale = FALSE`, `main = TRUE`; PyMC's float32 $\log 2$	kept / documented
D104–D107	grouping terms	`mm()` slopes and `scale = FALSE`, `by` with slopes, IDs within one formula	kept

5 How the claims were checked, and their limits

- Each feature: the gate tool on the tree that merged; the full non-sampling suite on it; the adapter and gate changes read for weakening (all extensions: `cs()` matching, shrinkage parameters, `hypothesis` and `conditional_effects` checks, G4 reading `expected` for every family with trials); every earlier case model unchanged (snapshots of `target_identity`, `describe()`, names and coefficients).
- Reproducibility: `oracle/regen.py check -all` regenerated all 134 frozen fixtures from R on `main` after the last merge and compared them with the committed files: all 134 identical. The four committed 50-digit references, the new cumulative probit one included, regenerate byte for byte (a harness test).

Limits I know of. Nothing in the loop samples, so no Tier D model has been fitted; the hosted sampling tests have not run on anything since C05. `conditional_effects()` is checked against brms on the two frozen cases only; on every other frozen model it takes it is checked for consistency with `predict()`, which G4 checks against brms. D06's names for brms's per-response, per-component and per-category coefficients, quoted in D99, were not probed.

6 Process and cost

Agent tokens reported by the agents that finished: D01 0.67M, D08 0.65M, D03 (the resumed agent) 0.59M, D06 0.41M, D05 0.35M, about 2.7M. The four agents stopped at your pause, and the fixtures agent, which died on a network error after finishing its work, do not report their usage. Finishing

D02, D04 and D07 myself, and doing every rebase myself, kept the rest of Tier D out of agent spend. The merge queue ran one suite at a time.

7 Decisions

1. **Reporting the brms bug.** Do you want D86 reported upstream (an issue on brms's GitHub with the reproducible R script)? It would go out under your account, so I will not post it unless you say so. I can draft it for you to post instead.
2. **What Tier E covers.** The candidates, roughly in order of value for the effort: (a) the simpler missing families (`zero_one_inflated_beta`, `zero_inflated_beta_binomial`, `hurdle_cumulative`, `inverse.gaussian`, `von_mises`, `asym_laplace`, `frechet`, `gen_extreme_value`, `xbeta`); (b) contexts refused today across families: distributional formulas, `cens()`/`trunc()` and `weights()` for the newer families, and their other links; (c) the multi-category families (`multinomial`, `dirichlet`, `dirichlet_multinomial`, `logistic_normal`); (d) post-processing: prior draws (`sample_prior`) and Savage–Dickey ratios, `kfold`, stacking weights, interaction effects; (e) the other correlation structures (`car`, `sar`, `fcor`, `unstr`, `cosy`); (f) `wiener` and `cox`, the hardest. Unless you say otherwise I would take (a), (b) and (d) next.
3. **Splines** (open since Tier B). My recommendation is unchanged: a clean-room rewrite of the spline module from Wood's published algorithms, keeping the tests against mgcv's output.
4. **CI.** Still refused for billing. Separately, GitHub notes that `ubuntu-latest` moves to Ubuntu 26 from 19 October.
5. **Attribution.** The ArviZ `NOTICE` question from Tier C is still open.
6. **A first real fit.** A sampling run of each Tier D family on the GPU or the cluster would test what no gate can. It needs your approval for the compute.